

# Mohammad Safaei



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## Education

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| 2022 – Present | <b>Ph.D. in Geography</b><br>Department of Geography, University of Florida, Gainesville, FL, USA<br>Cumulative GPA: <b>4/4</b><br>Expected graduation: <b>May 2027</b><br><b>Dissertation:</b> Data-Driven Conservation: Integrating AI and Remote Sensing for Monitoring Ecosystem Changes |
| 2016 – 2019    | <b>M.Sc. in Remote Sensing and Geographic Information System (GIS)</b><br>Department of Remote Sensing and GIS, Kharazmi University, Tehran, Iran<br>Overall GPA: <b>19.30/20 (4/4)</b> (Ranked 1 <sup>st</sup> )                                                                            |
| 2012 – 2016    | <b>B.Sc. in Natural Resources Engineering</b><br>Yazd University, Yazd, Iran<br>Overall GPA: <b>17.70/20 (3.67/4)</b> (Ranked 1 <sup>st</sup> )                                                                                                                                              |

## Graduate Certificates

| Mapping with Small Unmanned Aerial Systems (UAS)                                                                                                                                                 | Digital Geography and GIS                                                                                                                                                  | Medical Geography                                                                                                                                                                                   |
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| <ul style="list-style-type: none"><li>• FAA Part 107 Remote Pilot Certificate</li><li>• Flight planning and deployment</li><li>• Field survey</li><li>• Photogrammetry</li><li>• LiDAR</li></ul> | <ul style="list-style-type: none"><li>• Geospatial data analysis</li><li>• Remote sensing image processing</li><li>• Geo-visualization</li><li>• GIS Programming</li></ul> | <ul style="list-style-type: none"><li>• GIS for zoonoses and disease ecology</li><li>• GIS models for public health</li><li>• Landscape epidemiology</li><li>• Disease diffusion modeling</li></ul> |

## Research Interests

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| <ul style="list-style-type: none"><li>• Remote sensing and geospatial analysis</li><li>• GeoAI and geospatial modeling</li><li>• UAS-based remote sensing and data collection</li></ul> | <ul style="list-style-type: none"><li>• Land-cover change detection</li><li>• Vegetation dynamics and time-series analysis</li><li>• Conservation and landscape ecology</li></ul> | <ul style="list-style-type: none"><li>• Disease mapping and spatial epidemiology</li><li>• Vector-borne disease and ecological niche modeling</li></ul> |
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## Technical Expertise

- **Programming for geospatial workflows:** Python, R, JavaScript
- **GeoAI:** Machine learning and deep learning for remote sensing and GIS applications using TensorFlow and PyTorch.
- **Cloud & geospatial computing:** Google Earth Engine and AWS
- **Web GIS and web mapping:** Interactive web applications and 3D geospatial visualizations using HTML, CSS, and web mapping libraries.

## Geospatial Software & Platforms

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| <ul style="list-style-type: none"><li>• Esri ArcGIS platform (ArcGIS Pro, ArcGIS Online, StoryMaps, Survey123)</li><li>• QGIS</li><li>• Pix4D</li><li>• PostgreSQL/PostGIS and SQL Server</li><li>• GeoDa</li></ul> | <ul style="list-style-type: none"><li>• SaTScan</li><li>• ENVI</li><li>• ERDAS IMAGINE</li><li>• Global Mapper</li><li>• PCI Geomatica</li></ul> |
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# Teaching Experience

**2022-Present** Instructor, Department of Geography, University of Florida.

**GIS 3043/5107C: Foundations of GIS / GIS in Research** (4 cr.):  
*Fall 2025-Summer 2026*

- Instructor for a combined undergraduate/graduate course introducing GIS concepts, spatial analysis, and geographic data visualization.
- Designed and delivered hands-on labs in thematic mapping, cartography, georeferencing and digitizing, spectral analysis, land-cover change analysis, spatial interpolation, and LiDAR data analysis.
- Guided students through independent final projects.

**GIS 2002: The Digital Earth** (3 cr.):  
*Fall 2022-Summer 2025*

- Instructor for an online undergraduate course introducing digital geographic data, maps, satellite imagery, and aerial photography.
- Taught foundational GIS and remote sensing concepts for visualizing and analyzing environmental and societal patterns.
- Guided students in creating maps and performing basic spatial analysis using digital geospatial data.

**Additional Courses Taught:** GEO 2301: The Next Pandemic (TA); GEO 3352: The Human Footprint on Landscape; GEO 3427: Plants, Health, and Spirituality; GEO 2500 Global and Regional Economies (TA).

## Publications (Total: 8; 1<sup>st</sup>-author: 3; h-index: 4; i10-index: 3)

- 2026** Southworth, J., **Safaei, M.**, Rahaman, M., Herrero, H., Smith, A. C., Tefera, B. B., ... & Alruzuq, A. R. (2026). GEOAI Methods for Land Use and Land Cover Classification: Machine Learning and Deep Learning Approaches. In *Remote Sensing, Big Data, and GeoAI* (pp. 187-219). Elsevier. ([link](#))
- 2026** Gibbes, C., Southworth, J., Bunting, E., Rahaman, **M.**, **Safaei, M.**, Chakraborty, T., ... & Smith, A. C. (2026). Foundations of Remote Sensing and Earth Observation. In *Remote Sensing, Big Data, and GeoAI* (pp. 43-90). Elsevier. ([link](#))
- 2025** Tefera, B. B., Southworth, J., Mossa, J., Rahaman, M., **Safaei, M.**, Yang, D., & Karuppannan, S. (2025). Predictive groundwater quality responses to land cover and lithology in the upper Awash River basin (Ethiopia) with stacking ensembles. *Journal of Environmental Management*, 394, 127572. ([link](#))
- 2025** **Safaei, M.**, Southworth, J., Gibbes, C., Herrero, H. V., Rahaman, M., Tefera, B. B., & Blackburn, J. K. (2025). Land-cover classification in Addo Elephant National Park: Analyzing the impact of variables, classifiers, and object-based approach. *Ecological Informatics*, 103279. ([link](#))
- 2025** Rahaman, M., Southworth, J., Amanambu, A. C., Tefera, B. B., Alruzuq, A. R., **Safaei, M.**, ... & Smith, A. C. (2025). Combining deep learning and machine learning techniques to track air pollution in relation to vegetation cover utilizing remotely sensed data. *Journal of Environmental Management*, 376, 124323. ([link](#))
- 2024** Southworth, J., Smith, A. C., **Safaei, M.**, Rahaman, M., Alruzuq, A., Tefera, B. B., ... & Herrero, H. V. (2024). Machine learning versus deep learning in land system science: a decision-making framework for effective land classification. *Frontiers in Remote Sensing*, 5, 1374862. ([link](#))
- 2022** **Safaei, M.**, Rezayan, H., Zeaiean, P., & Torahi, A. (2022). Modelling potential impacts of climate change on the oak spatial distribution (Case study: Ilam and Lorestan provinces). *Journal of Applied researches in Geographical Sciences*. ([link](#))
- 2021** **Safaei, M.**, Rezayan, H., Zeaiean, P., & Sadidi, J. (2021). Optimization of species distribution models using a genetic algorithm for simulating climate change effects on Zagros forests in Iran. *Ecological Informatics*, 63, 101288. ([link](#))

## Manuscripts Under Review and in Preparation

Tefera, B. B., Southworth, J., Yang, D., Kiker, G. A., **Safaei, M.**, et al. Long-term terrestrial water-storage dynamics and groundwater stress in Stellenbosch, South Africa, 2002–2024. *Journal of Environmental Management*. **Under review**.

**Safaei, M.**, & Southworth, J. Subpixel Land-Cover Change Monitoring Using Spectral Unmixing and Trend Detection. *Manuscript in preparation*.

**Safaei, M.**, & Blackburn, J. K. Joint Species Distribution Modeling of *Francisella tularensis*–Associated Hosts and Vectors Across Ukraine. *Manuscript in preparation*.

## Conferences and Professional Workshops

- **Ecology and Evolution of Infectious Diseases (EEID) Conference (2026)**, Blacksburg, VA.
- **VectorByte Training Workshop (2026)**, Blacksburg, VA.
- **Esri User Conference (2026)**, San Diego, CA.
- **Geo Week (2026), Denver, CO — Poster:** Subpixel Monitoring of Land-Cover Changes Using Spectral Mixture Analysis and Trend Analysis Techniques.
- **American Association of Geographers (AAG, 2026), San Francisco, CA — Paper:** Time Series Subpixel Mapping of Land-Cover Change with MESMA and BEAST Algorithms.
- **UF/FL-ASPRS Fall Geospatial Workshop (2025), Apopka, FL — Paper:** Quantifying Canopy Structure Change at NEON TALL with Discrete-Return LiDAR (2016–2024).
- **Southeastern Division of the American Association of Geographers (SEDAAG, 2025), Lexington, KY — Poster:** Continuous vs. Discrete Approaches to Land Cover Change Detection.
- **AAG (2025), Detroit, MI — Poster:** Assessing Land-Cover Changes in Addo Elephant National Park Using Continuous and Discrete Approaches.
- **AAG (2024), Honolulu, HI — Paper:** Land Use and Land Cover Classification in Addo Elephant National Park: Analyzing the Impact of Variables, Algorithms, and Object-Based Approach.
- **Florida Society of Geographers (FSG, 2024), St. Augustine, FL — Poster:** Object-Based Classification for Land Cover Change Mapping in Addo Elephant National Park Using Landsat Imagery.
- **SEDAAG (2023), Norfolk, VA — Poster:** Precision Land Use and Land Cover Mapping in Addo Elephant National Park Through Landsat-8 and Google Earth Engine.
- **SEDAAG (2022), Atlanta, GA — Poster:** Optimization of Species Distribution Models Using a Genetic Algorithm for Simulating Climate Change Effects on Zagros Forests in Iran.

## Leadership and Service

- **Vice President, University of Florida Student Chapter of ASPRS** (American Society for Photogrammetry and Remote Sensing) 2025 – Present
- **Representative, UF Geography Student Services Committee (SSC)**, 2026 – 2027
- **Peer Reviewer:** *Discover Environment*, 2026
- **Peer Reviewer:** *Modeling Earth Systems and Environment*, 2025
- **Member of Scientific Association, Department of Natural Resources Engineering, Yazd University**, 2013 – 2016

## Honors and Awards

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- **International Geographic Information Fund (IGIF) Graduate Research Award**, American Association of Geographers (AAG), 2026
- **Top Published Student Research Article Award**, Evening of Excellence, Department of Geography, University of Florida, 2026
- **Excellence in Teaching Award**, University of Florida, Spring 2024 and Fall 2024
- **Poster Award**, UF/FL-ASPRS Fall Geospatial Workshop, 2025
- **Ranked 1st in M.Sc. Cohort**, Kharazmi University, Tehran, Iran, 2019
- **Ranked 1st in B.Sc. Cohort**, Yazd University, Yazd, Iran, 2016

## Travel Grants and Awards

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- **Evan Coe Travel Award in Medical Geography** (\$2,000), 2026
- **Florida Region ASPRS Andregg-Beasley Student Travel Award** (\$650), 2026
- **CLAS Graduate Student Travel Fund** (\$450), 2026
- **UF Research Travel Award** (\$400), 2026
- **Geography Graduate Student Travel Award** (\$400), 2026
- **Graduate Student Council Travel Grant** (\$400), 2026
- **UF Research Travel Award** (\$300), 2025
- **Geography Graduate Student Travel Award** (\$300), 2025
- **Graduate Student Council Travel Grant** (\$350), 2024
- **Geography Graduate Student Travel Award** (\$300), 2024
- **Geography Graduate Student Travel Award** (\$300), 2023

## References

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- Dr. Jane Southworth (Ph.D. advisor), Professor, Department of Geography, University of Florida  
Email: [jsouthwo@ufl.edu](mailto:jsouthwo@ufl.edu)
- Dr. Jason Blackburn, Professor and Chair, Department of Geography, University of Florida  
Email: [jblackburn@ufl.edu](mailto:jblackburn@ufl.edu)
- Dr. Di Yang (Committee member), Assistant Professor, Department of Geography, University of Florida  
Email: [yangdi1031@ufl.edu](mailto:yangdi1031@ufl.edu)